

READ THIS FIRST

The formula sheet gives you the quadratic formula, the binomial expansion, the trig identities and the sine/cosine rules. **Everything below is NOT on it.** If you only have twenty minutes, learn the red boxes — they are where real marks were lost in marked scripts, not where the hard questions were.

DISCRIMINANT — THE WHOLE OF IT

$$D = b^2 - 4ac$$

$D > 0$ 2 distinct real roots; line cuts curve twice

$D = 0$ equal roots; line is a **tangent**

$D < 0$ no real roots; **no intersection**

"Always positive" (curve above the x -axis for all x): **$a > 0$ AND $D < 0$.**

"Always negative": **$a < 0$ AND $D < 0$.**

Curve above a line? Subtract first: (curve – line) > 0 for all x , then apply the rule to *that* quadratic.

TRAP — SOLVING IN K

$k(k - 12) < 0$ gives **$0 < k < 12$** (inside).

$k(k - 12) > 0$ gives **$k < 0$ or $k > 12$** (outside).

Sketch the parabola every single time. Below the axis = between the roots. This one line of drawing has saved 4 marks more than once.

SURDS

Conjugate of $a + \sqrt{b}$ is $a - \sqrt{b}$ — **flip the sign of the surd only.**

Denominator $2\sqrt{3} - 3 \rightarrow$ multiply by $2\sqrt{3} + 3$, giving $12 - 9 = 3$.

Rational $\neq$ integer. If it asks for $a + b\sqrt{3}$ with a, b rational, a fraction like $11/6$ is the expected answer — do not round it.

Surd equations: isolate one surd, square, isolate again if needed, square again. **Then test every root in the ORIGINAL equation.** Squaring invents roots. Sometimes both are valid — do not reject one out of habit.

POLYNOMIALS

Factor theorem: $(x - a)$ is a factor $\Leftrightarrow f(a) = 0$.

For $(2x - 1)$, use $f(1/2)$, not $f(2)$.

Remainder theorem: remainder on dividing by $(x - a)$ is $f(a)$.

Two conditions $\rightarrow$ two simultaneous equations in the two unknowns.

"**Only one real root**": factor out the linear part, then show the quadratic factor has **$D < 0$** . Say it in words to earn the mark.

PARTIAL FRACTIONS

$$(ax + b) \frac{A}{ax+b}$$

$$(ax + b)^2 \frac{A}{ax+b} + \frac{B}{(ax+b)^2}$$

$$(x^2 + c) \text{ irreducible } \frac{Ax+B}{x^2+c}$$

Improper first. If the numerator's degree $\geq$ the denominator's, divide out a quotient BEFORE splitting. Degree 3 over degree 3 gives a constant quotient.

If the denominator is given unfactorised, factorise by grouping first: $x^3 - x^2 + 4x - 4 = x^2(x-1) + 4(x-1)$.

TRAP — LOGARITHM LAWS

$$\log A + \log B = \log(AB) \text{ — NOT } \log(A+B)$$

$$\log A - \log B = \log(A/B) \text{ — NOT } \log(A-B)$$

$$n \log A = \log A^n$$

$$\text{Change of base: } \log_b x = \frac{\log_a x}{\log_a b} \cdot \log_x a = \frac{1}{\log_a x}$$

$$\log_9 x = \frac{1}{2} \log_3 x \cdot \log_4 x = \frac{1}{2} \log_2 x$$

Always check the domain. Every argument must be > 0 , so reject roots that make any bracket zero or negative. The base must be > 0 and $\neq 1$.

INDICES & EXPONENTIALS

Hidden quadratic: let $u = a^x$.

Watch the stray factor: $3^{2x+1} = 3(3^x)^2$, not $(3^x)^2$.

a^x is always positive — reject any negative value of u , and say why.

Growth/decay $M = M_0 e^{kt}$: two readings $\rightarrow$ **divide the equations** to kill M_0 , then substitute back for M_0 . If neither reading is at $t = 0$, M_0 is NOT the first mass given.

In M_0 is not M_0 . Undo the logarithm at the end.

BINOMIAL THEOREM

$$\text{General term: } T_{r+1} = {}^nC_r a^{n-r} b^r$$

In $(2 + kx)^6$, the x^3 coefficient is ${}^6C_3 2^3 k^3$ — **keep BOTH the power of 2 and the power of k.** Dropping the 2^{n-r} is the single commonest error.

Term independent of x = the x^0 term: set the total power to zero and solve for r .

Bracket in front? $(2 - x)(1 + 2x)^8$: the x^5 term takes **two** contributions — $2 \times (x^5 \text{ term})$ and $(-x) \times (x^4 \text{ term})$.

COORDINATE GEOMETRY

$$\text{Gradient } (y_2 - y_1) / (x_2 - x_1)$$

$$\text{Perpendicular } m_1 m_2 = -1$$

$$\text{Midpoint } ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

$$\text{Distance } \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

$$\text{Area (shoelace)} \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Perpendicular bisector = midpoint + negative reciprocal gradient. Every point on it is **equidistant from both ends** — that one sentence proves a triangle isosceles.

"Solutions by accurate drawing will not be accepted" means show algebra. A measured answer scores zero even if it is right.

LINEAR LAW

Rearrange into **$Y = mX + c$** , then say plainly what is plotted against what.

$$y = ax^b \quad \lg y \text{ against } \lg x \text{ — log both}$$

$$y = ab^x \quad \lg y \text{ against } x$$

$$y = x/(ax+b) \quad 1/y \text{ against } 1/x, \text{ or } x/y \text{ against } x$$

Read gradient and intercept off the DRAWN LINE, using two points far apart. Never go back to the raw table. If the intercept is $\lg a$, you still need **$a = 10^{\text{intercept}}$** .

TRAP — TRIG SIGNS

ASTC from quadrant 1: All $\cdot$ Sin $\cdot$ Tan $\cdot$ Cos are positive. **$\sqrt{1 - p^2}$ is NEVER $1 - p$.** Test it: $p = 0.6$ gives 0.8 , not 0.4 . A root does not distribute over a subtraction.

$$\sin(90^\circ \pm A) = \cos A$$

$$\cos(90^\circ + A) = -\sin A$$

$$\sin(180^\circ - A) = \sin A$$

$$\cos(180^\circ - A) = -\cos A$$

$$\tan(180^\circ - A) = -\tan A$$

$$\text{Any } (360^\circ - A) \quad \text{quadrant 4: only cos stays +}$$

sec, cosec, cot: find $\sin / \cos / \tan$ first, then **take the reciprocal LAST**.

TRIG GRAPHS & EQUATIONS

For **$y = a \pm b \sin(nx)$** : amplitude **b** , period **$360^\circ/n$** (or $2\pi/n$), max **$a + b$** , min **$a - b$** .

A minus sign swaps where max and min happen — $y = 3 - 4\cos(x/2)$ starts at its MINIMUM.

Before sketching, ask: **how many cycles fit the given range?** Period 4π over 0 to 4π is exactly one.

Solving: transform the RANGE too. If $0 \leq x \leq 4\pi$ and the angle is $x/2$, then work in $0 \leq x/2 \leq 2\pi$ and double at the very end.

Negative cosine $\rightarrow$ **quadrants 2 and 3**, so $180^\circ - \alpha$ and $180^\circ + \alpha$. Negative sine $\rightarrow$ **quadrants 3 and 4**. Getting this backwards costs both accuracy marks.

Proving an identity: work on ONE side only. Cross-multiplying both sides assumes the result and scores nothing.

MARKS YOU CAN BANK TOMORROW

- **Answer the form asked for.** "Exact" means surds, π or $\ln$ — never a decimal. "3 s.f.", "2 d.p.", "as a multiple of π ", "integers", "rational" — a right number in the wrong form loses the accuracy mark.
- **Ceiling answers.** "Least number of complete hours/days" — 8.94 becomes **9**, not 8. Say the quantity first passes the target during that unit.
- **"Explain why" is worth ~2 marks a paper.** Write a sentence, not just algebra.
- **"Hence"** means use the previous part. Starting again wastes time and may score zero.
- **Never leave a part blank.** Write the general term, the formula, the first substitution. Method marks are given for the attempt.
- **Keep unrounded values in the calculator** for the next part; round only the final answer.
- **Re-read the last line against the question.** Most marks in marked scripts were lost on the FINAL step of a correct solution — the wrong variable reported, the wrong form, a copied-down slip.

TONIGHT & TOMORROW

Tonight: read the red boxes twice. Do not start a new topic — you will not learn it in one evening and it will cost you sleep, which is worth more marks. Pack your calculator (check the batteries), a ruler, a sharp pencil for graphs, and your ID.

In the exam: ~1.5 minutes per mark. A 12-mark question deserves 18 minutes; a 4-mark opener should take 6. If a part is stuck after two minutes, move on and come back — later parts are often independent.

Last five minutes: check every answer has the units, the form and the sign the question asked for. That sweep is worth more than one more attempt at a hard part.